

- 4 A light-dependent resistor (LDR) is connected across the terminals of a cell that has internal resistance. A voltmeter is connected across the LDR.

(a) Draw a circuit diagram to show this arrangement.

[3]

(b) The cell has an electromotive force (e.m.f.) of 4.90 V.

At a particular light intensity, the LDR has a resistance of $230\ \Omega$ and a potential difference (p.d.) across it of 4.78 V.

- (i) Identify **one** possible source of systematic error in measuring the p.d. across the LDR using a digital voltmeter.

.....
..... [1]

(ii) Calculate the power dissipated in the LDR.

power = W [2]

(iii) Determine the internal resistance of the cell.

internal resistance = Ω [2]

- (iv) The intensity of the light illuminating the LDR is gradually decreased.

By placing **one** tick (✓) in each row, complete Table 4.1 to show what happens to the quantities indicated as the light intensity decreases.

Table 4.1

	decreases	unchanged	increases
resistance of LDR			
current in cell			
p.d. across LDR			

[3]

[Total: 11]